6.1 Altered Level of Consciousness

An alteration in mental status is abnormal and warrants examination.

NOTICE

Check a bedside blood glucose and treat hypoglycemia straight away

6.1.1 AEIOU TIPS

The mnemonic AEIOU TIPS may be used to systemically consider likely causes of mental status changes. Use the applicable condition specific guideline to treat identified conditions.

A	Alcohol, AMI, Acidosis
E	Epilepsy, Electrolyte, Endocrine, Environment
I	Insulin
O	Oxygen, Overdose
U	Uremia
T	Trauma, Temperature
I	Infection
P	Psychiatric, Pulmonary
S	Stroke, Shock

6.1.2 Glasgow Coma Scale

The GCS should be used to obtain, track, and communicate the mental status and level of consciousness on all patients with acute brain injury and other causes of altered mental status. The score allows for ongoing monitoring of important trends in the condition of a patient. A structured approach with an emphasis on independently assessing and reporting the three components should be used.

GCS Assessment Procedure

1. Check for factors interfering with communication, ability to respond and other injuries
2. Observe eye-opening, content of speech and movements of right and left sides
3. Depending on the patient's age and condition, stimulate the patient by using sound or physical stimuli. For sound, use a spoken or shouted request. For physical, use pressure applied to the fingertip, trapezius or supraorbital notch.
4. Rate each component based on the highest response observed.

Assessment of Non-verbal Children

The Pediatric Glasgow Coma Scale should be used to track and communicate the mental status and level of consciousness in pre-verbal children aged ≤ 2 years.

6.1.3 GCS

Criterion	Rating	Score
Eye Opening		
Open before stimulus	Spontaneous	4
After spoken or shouted request	To sound	3
After fingertip stimulus	To pressure	2
No opening at any time, no interfering factor	None	1
Closed by local factor	Non-testable	NT
Verbal Response		
Correctly gives name, place and date (Ped—Coos, babbles)	Orientated	5
Not orientated but communicating coherently (Ped—Irritable cries)	Confused	4
Intelligible single words (Ped—cries to physical stimulus)	Words	3
Only moans / groans	Sounds	2
No audible response, no interfering factor	None	1
Factor interfering with communication	Non-testable	NT
Best Motor Response		
Obey 2-part request (Ped—moves purposefully)	Obeys command	6
Brings hand above clavicle to stimulus on head neck	Localizing	5
Bends arm at elbow rapidly but features are not predominantly abnormal	Normal flexion	4
Bends arm at elbow, features clearly predominantly abnormal	Abnormal flexion	3
Extends arm at elbow	Extension	2
No movement in arms / legs, no interfering factor	None	1
Paralyzed or other limiting factor	Non-testable	NT

NOTE: **Pediatric criterion modifiers for children aged ≤ 2 years are located parenthetically in the GCS table.**

6.2 Behavior Disturbances and Agitation

Dangerous agitation in patients during transport poses significant risks to both the crew and the patient. Agitation can lead to unpredictable and violent behaviors, compromising the safety of the flight by distracting or physically harming the crew, interfering with medical care, or destabilizing the aircraft. Anticipating and managing agitation swiftly and effectively is crucial to ensuring the safety of patients and crew.

6.2.1 Assessment

An assessment of the patient's mental status, presence of potential triggers for agitation and effective intervention is crucial in providing effective care and preventing escalation.

Prior to transport, the patient needs to be formally assessed for anxiety/risk for agitation.

This assessment should include evaluation of:

- Prior violent behavior
- Cognitive Impairment
- Preexisting Agitation
- Drug or Alcohol effect or withdrawal
- Verbal abuse

Mental Status

- Assess the patient's orientation to person, place and time.
- Evaluate cognitive functions, including memory, attention, and simple problem-solving abilities.
- Observe the patient's mood, affect, and overall demeanor.
- Identify any signs of delirium, psychosis, or other psychiatric conditions.
- Conduct a thorough history of any known mental health disorders or substance use.

Agitation Triggers

- **Environmental Factors**: Overstimulation from noise, lights, confinement and crowding are ubiquitous to the transport environment.
- **Medical Conditions**: Acute medical issues such as infections, hypoxia, dehydration, or pain; chronic conditions like dementia or traumatic brain injury; withdrawal from alcohol or drugs.
- **Psychiatric Conditions**: Anxiety disorders, bipolar disorder, schizophrenia, or depression; history of violent behavior or self-harm.
- **Substance Use**: Intoxication or withdrawal from alcohol, drugs, or medications; interactions between substances and prescribed medications.
- **Communication Barriers**: Language differences, sensory impairments or cognitive impairments that hinder effective communication.
- **Personal Stressors**: Recent loss, financial difficulties, relationship problems, or other significant life changes.

Risk Assessment

A transport risk assessment should be completed on all patients with a history of behavior that may be a hazard to crew or patient. The following may indicate a higher risk for behavior disturbances and agitation in transport risk.

- History of violence or combative behavior prior to the Medical Crew's arrival.
- Expressions of anger, frustration or agitation.
- Signs of intoxication or known history of substance abuse.

NOTICE Pharmacological management **and** physical restraints **may be necessary to ensure safe transport.**

6.2.2 Sedation Scales

The SAS or RASS should be used to assess and reassess agitated patients.

Riker Sedation-Agitation Scale (SAS)

Score	Term	Descriptor
7	Dangerous Agitation	Pulling at ET tube, trying to get off cot, striking at crew, thrashing side-to-side
6	Very Agitated	Requiring restraint and frequent verbal reminding of limits, biting ETT
5	Agitated	Anxious or physically agitated, calms to verbal instructions
4	Calm and Cooperative	Calm, easily arousable, follows commands
3	Sedated	Difficult to arouse but awakens to verbal or physical stimuli, follows commands but drifts off
2	Very Sedated	Arouses to physical stimuli but does not communicate or follow commands, may move spontaneously
1	Unarousable	Minimal or no response to noxious stimuli, does not communicate or follow commands

Richmond Agitation Sedation Score (RASS)

Score	Term	Description
4	Combative	Combative or violent, immediate danger to staff
3	Very agitated	Pulls on or removing catheters/tubes or aggressive behavior
2	Agitated	Frequent nonpurposeful movement or ventilator dyssynchrony
1	Restless	Anxious but movements not aggressive or vigorous
0	Alert and calm	
-1	Drowsy	Not fully alert, >10 secs awakening, with eye contact, to voice
-2	Light sedation	Briefly (< 10 secs) awakens with eye contact to voice
-3	Moderate sedation	Any movement (but no eye contact) to voice
-4	Deep sedation	No response to voice, any movement to physical stimulation
-5	Unarousable	No response to voice or physical stimulation

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6.3 At Risk for Dangerous Agitation

Indications

- Behavior that interferes with crew duties and therapies
- Physical evidence of anxiety (e.g. restlessness, panic)
- Intoxication (alcohol or drugs)
- Withdrawal symptoms from substances
- History of anxiety or delirium
- Other concerning behaviors or signs

Management

- Treat prior to loading in the aircraft if possible.
- Consider notifying the pilot of risk

Further Preparation

Anxious patients and patients at risk for Dangerous Agitation can escalate to Dangerous Agitation at any time. Therefore preparations should be made to respond to Dangerous Agitation as Follows:

Preparations required at Head Of Bed:

- Appropriately sized I-gel
- Appropriately sized BVM kit
- Appropriate medications
- Ensure availability of all airway management equipment and appropriate personnel

NOTE: **Airway management must be performed if patient's ventilatory status becomes compromised and is mandatory with administration of neuromuscular blockade.**

Droperidol—At Risk for Dangerous Agitation

Preferred initial therapy for adults.

Adult

- 2.5 mg IV, IM, or IO
- Repeat in 3 minutes if needed
- Maximum 10 mg

NOTE: **Maximum effects may not occur for 20–30 minutes**

NOTE: **May be used as monotherapy or in combination with a benzodiazepine**

NOTE: **Consider a 50% dose reduction in elderly patients or those with significant comorbidities to reduce the risk of respiratory depression and prolonged sedation.**

NOTE: **Haloperidol may be used as an alternative agent to Droperidol.**

WARNING **Ketamine may not work as well in the agitated alcohol withdrawal patient or the acutely stimulant intoxicated patient. In these patients, every 5 minute dosing of benzodiazepines with coadministration of an antipsychotic is preferred.**

WARNING **There are no pediatric indications or standing orders for At Risk Dangerous Agitation. Administration requires a physician order.**

Midazolam—At Risk for Dangerous Agitation

Adult

- 5 mg IV, IM, or IO
- Repeat in 5 min if needed

NOTE: Droperidol is first line treatment for At Risk Dangerous Agitation. Midazolam co-therapy may be considered.

NOTE: Consider a 50% dose reduction in elderly patients or those with significant comorbidities to reduce the risk of respiratory depression and prolonged sedation.

6.3.1 Dangerous Agitation

Anxiety or Agitation of any type may progress rapidly to *Dangerous Agitation* The safety of the crew, the flight, and the patient is paramount. Chemical restraint and patient control using dissociative doses of ketamine intramuscularly or intravenously may be used to rapidly control patients with dangerous agitation and combativeness that puts the crew or patient at risk for harm.

WARNING **With onset of *Dangerous Agitation*→ DECLARE A CLINICAL MAYDAY—NOTIFY PILOT**

- Sudden unanticipated agitation that cannot be controlled or becomes a risk to safety in transport should prompt the medical crew to request the pilot land or driver stop immediately.
- Sedation using a disassociative dose of ketamine should be used to rapidly control dangerous agitation in transport.

Further Preparation

Anxious patients and patients at risk for Dangerous Agitation can escalate to Dangerous Agitation at any time. Therefore preparations should be made to respond to Dangerous Agitation as Follows:

Preparations required at Head Of Bed:

- Appropriately sized I-gel
- Appropriately sized BVM kit
- Appropriate medications
- Ensure availability of all airway management equipment and appropriate personnel

NOTE: **Airway management must be performed if patient's ventilatory status becomes compromised and is mandatory with administration of neuromuscular blockade.**

6.3.2 Chemical Restraint

Chemical restraint and patient control using dissociative doses of ketamine intramuscularly or intravenously may be used to rapidly control patients with Dangerous Agitation and combativeness that puts the crew or patient at risk of harm.

NOTICE **Pulse oximetry (SpO2), cardiac rhythm, noninvasive blood pressure (NIBP), continuous waveform capnography and respiratory rate monitoring should be established as soon as practicable.**

Ketamine—Dangerous Agitation

Adult

- 3 mg/kg IBW IV/IO push or
- 4 mg/kg IBW IM
  - Use most concentrated form available
- Immediately follow IV/IO/IM administration with a Ketamine infusion 4 mg/kg/hr

Pediatric

- 3 mg/kg IBW IV/IO push or
- 4 mg/kg IBW IM
  - Use most concentrated form available
- Immediately follow IV/IO/IM administration with a Ketamine infusion 2 mg/kg/hr

NOTE: **Prepare for airway management and ventilation prior to initiating therapy**

NOTE: **Treat with ondansetron as per the Nausea and Vomiting Guideline when safe and practicable**

NOTE: **The full calculated dose of ketamine should be administered to control agitated uncooperative patients regardless of clinical problem, comorbidities, hemodynamic status, or presence of intoxicants.**

NOTE: **When utilizing Ketamine consider the usage of Rocuronium or Succinylcholine**

Rocuronium—Dangerous Agitation

Adult

- 1.5 mg/kg IBW IV, IO or
- 2 mg/kg IM (onset 5 min)

Pediatric

- No PHI Standing Order Or Indication

Succinylcholine—Dangerous Agitation

Adult

- 4 mg/kg IM (onset 3-4 minutes)

Pediatric

- No PHI Standing Orders or Indications

6.3.3 Patient Restraint

Patients who are potentially violent or may unintentionally strike out and harm themselves or others, may be restrained to protect the patient and providers, preventing injury or delay in treatment, where verbal deescalation techniques are unsuccessful or infeasible.

Restraint Application

- Medical crewmembers may apply restraints when necessary in accordance with these guidelines.
- Where practicable, medical control should be consulted when patient restraint is necessary.

NOTICE **When restraints are necessary, the procedure should be done in a way that protects the patient's health, safety, dignity, rights and well-being.**

- The method of restraint should be individualized to use the least restrictive method of restraint that protects the patient, medical crew, pilot and others from harm.

Indications

- Reasonable risk of violent action or unintentionally striking out
- Safe access for medical procedures when involuntary patient-interference or resistance is reasonably anticipated
- Prophylactic restraint before the patient becomes involuntarily resistant to medically necessary treatment
- Recovery from medication effects or anticipation of improved patient condition producing combativeness
- Evaluation or treatment of combative persons when illness or trauma is suspected to be the cause of combativeness
- Involuntary treatment of persons incompetent to refuse treatment

NOTICE **Verbal management techniques should be used prior to and during application of any restraints and, to the extent possible, providers should inform the patient that restraints will be applied.**

Physical Restraint Devices and Techniques

- Supplemental litter straps may be placed over the pelvis and lower extremities to reduce movement
- Commercially available extremity restraints made of padded leather or fabric may be used to prevent extremity movement

NOTICE **All methods of restraint must allow for quick release of the device, full access for patient care, airway control, and monitoring of patient's vital signs.**

Refusal to Transport and Alternate Modes of Transport

Air medical transport should be refused if the patients can not be adequately controlled or restrained to ensure the safety of all personnel during flight. The medical crew may use ground transportation if that reduces the risk of transport to an acceptable level.

Cross reference		
Clinical Care Guidelines	23	1-33 - Patient Restraint

6.4 Acute Coronary Syndrome

Advisory that care on an inter-facility transport should be collaborative between the referring and receiving providers, specifically in the instance of thrombolytic/anticoagulant therapy

6.4.1 Assessment

- Multi-function electrodes should be applied in anticipation of life-threatening arrhythmias.
- AED pads should be placed empirically with anterior-posterior placement as the preferred site compared to anterior-lateral

6.4.2 12 Lead ECG

Obtain or review a recent 12 lead ECG and determine the:

- Heart Rate
- Cardiac rhythm
- Axis
- Presence and localization of injury, ischemia, or infarction
- Obtain a right-sided 12 lead ECG to confirm diagnosis of RVMI

Posterior 12-Lead ECG

When

Posterior ECG may be performed on patients who present with signs of ischemia in inferior and or lateral leads, ST depression and/or large T waves in the anterior leads, larger R-waves in V2-V3 compared to V4-V6. Prompt management to intervention is the priority over non-standard ECG.

V7 – Left posterior axillary line, in the same horizontal plane as V6.

V8 – Tip of the left scapula, in the same horizontal plane as V6.

V9 – Left paraspinal region, in the same horizontal plane as V6.

Graphic

TBD

Right Sided 12-Lead ECG

When

Right Sided ECG utilizing V4R method should be performed on patients who present with an inferior wall MI.

- Obtain V4R at 5th right ICS mid-clavicular line
- ST elevation in V4R has high sensitivity and specificity for diagnosis of RVMI

Graphic

TBD

References

- Lizzo JM, Chowdhury YS. Posterior Myocardial Infarction. [Updated 2023 Aug 8]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2024 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK553168/
- Warner MJ, Tivakaran VS. Inferior Myocardial Infarction. [Updated 2023 Feb 12]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2024 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK470572/

6.4.3 Sgarbossa Criteria

The Sgarbossa Criteria are used to diagnose acute myocardial infarction (MI) in patients with a left bundle branch block (LBBB) or a paced rhythm. These criteria help distinguish between ST-segment changes caused by the conduction abnormality and those indicative of an infarction.

Sgarbossa Criteria

Concordant ST-Segment Elevation

- ST-segment elevation ≥ 1 mm in any lead with a positive QRS complex (concordant with QRS direction).
- Score: 5 points

Concordant ST-Segment Depression

- ST-segment depression ≥ 1 mm in leads V1, V2, or V3.
- Score: 3 points

Discordant ST-Segment Elevation

- ST-segment elevation ≥ 5 mm in any lead with a negative QRS complex (discordant with QRS direction).
- Score: 2 points

Diagnostic Criteria

- A total score of 3 or more points indicates a high probability of acute MI.

Sgarbossa Criteria Quick Reference Table

Criterion	ECG Finding	Score
Concordant ST-Segment Elevation	ST elevation ≥1 mm in leads with positive QRS	5
Concordant ST-Segment Depression	ST depression ≥1 mm in V1, V2, or V3	3
Discordant ST-Segment Elevation	ST elevation ≥5 mm in leads with negative QRS	2

Application

1. **Evaluate** the ECG using the Sgarbossa Criteria.
2. **Score** the findings based on the table.
3. **Assess** the total score for the probability of MI
4. **3 or more points**: High probability of acute MI

Notes

- Use the Sgarbossa Criteria in conjunction with clinical findings and patient history.
- Always consider obtaining online medical direction for ambiguous or borderline cases.

Reference

- Sgarbossa EB, Pinski SL, Barbagelata A, et al. Electrocardiographic diagnosis of evolving acute myocardial infarction in the presence of left bundle-branch block. N Engl J Med. 1996;334(8):481-487

Sgarbossa Criteria Quick Reference Table

Criterion	ECG Finding	Score
Concordant ST-Segment Elevation	ST elevation ≥1 mm in leads with positive QRS	5
Concordant ST-Segment Depression	ST depression ≥1 mm in V1, V2, or V3	3
Discordant ST-Segment Elevation	ST elevation ≥5 mm in leads with negative QRS	2

Sgarbossa Criteria Graphic

6.4.4 STEMI

Patients with ST elevation that meets the following criteria must be treated in a efficient and timely manner consistent with local practice and heath system expectations.

- ST elevaton = or > 0.1 mV in 2 or more contiguous leads
- New LBBB

6.4.5 Treatment

Aspirin

Adult

- 162–324 mg (2-4 pediatric chewable tabs)

Oxygen

Adult

- If SpO2 is < 90%
- Use cannula or mask and titrate to maintain SpO2 90–95%

Reperfusion arrhythmias

- Reperfusion arrhythmias should be managed with standard anti-arrhythmic measures. Precipitating or persistent ventricular fibrillation or tachycardia may be reflective of ongoing occlusion.

Normal Saline

- If patient presents with SBP < 110 mmHg, history and exam suggestive of inferior wall MI with suspected/known RVMI, administer fluid bolus 100 – 500 mL and repeat as needed to maintain vital signs providing that lung sounds remain clear.

Nitroglycerin

Administer nitroglycerin SL or IV for chest pain PRN

Do we need guidance for the continuation of NTG Paste?

Nitroglycerin—SL

Adult

- 0.4 mg SL
- Repeat PRN to desired effect

Ped

- No PHI Standing Orders or Indication

Nitroglycerin—Infusion

Adult

- 40–200 mcg/min IV via infusion pump
- Titrate 20–50 mcg/min every 3-5 min to desired effect

Ped

- No PHI Standing Orders or Indication

Nitroglycerin—Advisories

⚠ CAUTION **Do not administer nitroglycerin to patients who have taken avanafil within 12 hours, sildenafil or vardenafil within 24 hours, or tadalafil within 48 hours. Doing so may cause a severe drop in blood pressure.**

- Use cautiously with patients who present with an acute inferior wall MI
- A right sided ECG should be performed on patients who present with an inferior wall MI
- Use cautiously with patients who are nitroglycerine naive
- Hypotension is more common in the elderly
- Bradycardia and hypotension usually respond to supine positioning and IV fluids

Heparin

Consensus for additional guidance to support bolus only for transport due to duration of the bolus making an infusion a moot point for transport if there is under 3 hours unless PTA infusion is going and can be continued without delay or in the absence of PTA infusion

Heparin

Adult

- 60 units/kg initial bolus
- Max 4,000 units
- 12 units/kg/hr continuous infusion
- Max infusion 1,000 units/hr

Pediatric

- No PHI Standing Orders or Indications

The following medications may be employed prior to arrival. Refer to drug profile for additional information.

- Alteplase
- Tenecteplase
- Clopidogrel
- Ticagrelor

6.4.6 Special Considerations

- In cases of inter-facility transport where patients have received fibrinolytics, prepare for complications associated with reperfusion.
- Assess and document neurological assessment throughout transport

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6.5 Congestive Heart Failure

6.5.1 Signs and Symptoms

- S3, S4
- Oliguria
- Tachycardia
- Tachypnea
- Rales
- Dyspnea

Causes of Acute Pulmonary Edema

- Worsening heart failure
- Coronary insufficiency
- Subendocardial infarction
- Transmural infarction
- Acute dysrhythmia
- Medication noncompliance
- Dietary indiscretion
- Valvular insufficiency

Dyspnea and Hypoxia

Non-invasive Ventilation should be utilized as soon as it is practicable.

NOTE: **Escalate to endotracheal intubation PRN**

Oxygen

Adult

- 2–25 L/min titrated to an SpO2 of 93–99%

Pediatric

- Refer to adult dosing

Nitroglycerin—Infusion

Adult

- 40–200 mcg/min IV via infusion pump
- Titrate 20–50 mcg/min every 3-5 min to desired effect

Ped

- No PHI Standing Orders or Indication

If IV infusion is not immediately available

Nitroglycerin—SL

Adult

- 0.4 mg SL
- Repeat PRN to desired effect

Ped

- No PHI Standing Orders or Indication

NOTE: **Treat poor perfusion and hypotension as per Shock & Cardiovascular Instability protocol**

6.6 Allergic Reaction and Anaphylaxis

Patient may have an allergic reaction as their primary problem or suffer it as the result of therapy. If an allergic reaction or anaphylaxis is suspected perform a focused assessment with attention to:

- Airway changes or difficulty
- Respiratory distress
- CV instability
- Urticaria
- Voice changes
- Wheezing
- Hypoxemia
- Level of consciousness
- Use of beta-blockers

Determine the disease severity by assessing each body system and then stratify treatment based on severity.

- Airway and breathing difficulty, cardiovascular instability, or involvement of more that 1 body system should be considered a severe reaction or anaphylaxis.
- Signs and symptoms limited to integumentary involvement may be considered a mild or moderate reaction.

6.6.1 Mild to Moderate Allergic Reaction

Patients with isolated urticaria or other skin manifestations may be treated as moderate allergic reactions.

Diphenhydramine

Adult

- 25–50 mg IV, IM
- Repeat every 4 hours PRN

Pediatric

- 1–2 mg/kg IV, IM
- Maximum single dose 50 mg
- Repeat every 4 hours PRN

Methylprednisolone

Adult

- 125 mg IV over 1 minute

Pediatric

- 2 mg/kg IBW IV over 1 minute
- Maximum single dose 125 mg

6.6.2 Severe Allergic Reaction and Anaphylaxis

NOTICE

Early recognition and treatment is important—delay in EPINEPHrine administration is associated with increased mortality and biphasic reaction

Epinephrine

Adult

- 0.5 mg IM
- 1 mg/mL concentration
- Repeat every 20 min PRN
- Alternately Adult EpiPen

Pediatric

- 0.01 mg/kg IM
- Max single dose 0.5 mg
- 1 mg/mL concentration
- Repeat every 20 min PRN
- Alternately Pediatric EpiPen

Normal Saline

- 20 mL/kg IV bolus over 10 to 20 min
- Maximum single bolus 1000 mL
- Continue management of hypotension as per the CV instability and Shock Guideline

Albuterol

Adult

- 5 mg in 3–5 mL via nebulizer
- Repeat every 20 minutes for 3 doses PRN

Alternately

- 4 to 8 (90 mcg) actuations via metered-dose inhaler
- Repeat every 20 minutes for up to 4 hours PRN
- Then every 1 to 4 hours PRN

Pediatric

- Refer to adult dosing

Diphenhydramine

Adult

- 25–50 mg IV, IM
- Repeat every 4 hours PRN

Pediatric

- 1–2 mg/kg IV, IM
- Maximum single dose 50 mg
- Repeat every 4 hours PRN

Methylprednisolone

Adult

- 125 mg IV over 1 minute

Pediatric

- 2 mg/kg IBW IV over 1 minute
- Maximum single dose 125 mg

Concomitant Use of Beta Blocker

Glucagon

Adult

- 1-5 mg IV over 2 minutes
- Follow with an IV infusion at 3 mg/hour via pump

Pediatric

- 30 to 150 mcg/kg IV over 2 minutes
- Max dose 1 mg
- Follow with an IV infusion at 70 mcg/kg/hour via pump

6.7 Asthma

Perform a focused assessment with attention to the following

- History of current illness
- Mentation
- Verbalization
- Presence and quality of wheezing
- Accessory muscle use
- Presence of air-trapping
- Work of breathing
- Respiratory rate

- Assess home and PTA asthma medications and factor into the treatment plan
- Determine the disease severity and then treat as indicated in this guideline

6.7.1 Mild or Moderate Asthma

Mild to moderate asthma is characterized by dyspnea, wheezing, mild tachycardia, and some accessory muscle use.

Albuterol

Adult

- 5 mg in 3–5 mL via nebulizer
- Repeat every 20 minutes for 3 doses PRN

Alternately

- 4 to 8 (90 mcg) actuations via metered-dose inhaler
- Repeat every 20 minutes for up to 4 hours PRN
- Then every 1 to 4 hours PRN

Pediatric

- Refer to adult dosing

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6.7.2 Worsening or Severe Disease

Severe asthma is characterized by worsening of symptoms after inhaled β agonists, increasing dyspnea, speaking 1 or 2 words at a time, accessory muscle use, marked tachycardia, audible wheezing or absent air moment and mentation changes.

NOTICE **Minimize medication interruptions by monitoring closely for treatment completion and initiating the next dose in a timely manner.**

Albuterol

Adult

- Continuous aerosol administration using undiluted solution
- Continue until there is improvement

Pediatric

- Refer to adult dosing

Ipratropium

Adult

- 0.5 mg in 2.5 mL via nebulizer
- Additional doses on-line

Pediatric

- 0.25 to 0.5 mg in 2.5 mL via nebulizer
- Additional doses on-line

Methylprednisolone

Adult

- 125 mg IV over 1 minute

Pediatric

- 2 mg/kg IBW IV over 1 minute
- Maximum single dose 125 mg

Magnesium

Adult

- 2–3 gm IV infusion over 15–20 minutes
- Repeat PRN x 1
- Max single dose 3 gm
- Max total dose 6 gm

Pediatric

- 25–50 mg/kg IV infusion over 20 minutes
- Repeat PRN x 1
- Max single dose 3 gm
- Max total dose 6 gm

6.7.3 Severe Asthma Refractory to Treatment

Parenteral Epinephrine

If patient presents in extremis or does not improve with aggressive use of inhaled β agonists, ipratropium bromide, magnesium and steroids IM EPINEPHrine should be administered.

Epinephrine

Adult

- 0.5 mg IM
- 1 mg/mL concentration
- Repeat every 20 minutes up to 3 doses

Pediatric

- 0.01 mg/kg IM
- Max single dose 0.5 mg
- 1 mg/mL concentration
- Repeat every 20 minutes up to 3 doses

NOTE: **For severe asthma refractory to first line treatment**

Terbutaline Infusion

NOTICE **If called upon to manage a terbutaline drip to treat severe asthma, defer to the sending or receiving physician with respect to titration. In the absence of such guidance use the dosing in this profile.**

Terbutaline

Adult

- 10 mcg/kg IV over 10 minutes, followed by infusion
- Begin infusion at 0.3–0.5 mcg/kg/min IV
- Increase 0.5 mcg/kg/min every 30 min PRN
- Max dose 3 mcg/kg/min

NOTE: **For severe asthma refractory to first line treatment**

NOTE: **Requires both loading dose and infusion**

Ped

- Refer to adult dosing

6.7.4 Asthma Requiring Intubation

Intubation and positive pressure ventilation of the asthmatic patient is a “last resort” maneuver. Invasive airway management should be avoided if possible and preceded by maximally aggressive use of inhaled β agonists, IM epinephrine, ipratropium bromide, magnesium, steroids and NIPPV. Only if these measures fail, should intubation and mechanical ventilation be considered.

- Intubation should be timed and conducted to allow for precise post intubation ventilator management as per the Respiratory Support Guideline.
- Aerosol medications should be continued and administered via in line nebulizer
- Ketamine should be used for RSI and post intubation sedation

Clinical Markers for Intubation and Impending Respiratory Failure

- Decreasing level of consciousness and confusion
- Weak breathing efforts or “silent chest”
- Hypoxemia
- Severe acidosis
- Normalizing respiratory alkalosis (PaCO2 rising to 40 mm/Hg)
- Hypercapnia (EtCO2 > 50 mm / Hg)
- Hypotension and cardiac arrhythmias

Ventilator Management of the Asthmatic

Principles of ventilation of the Asthmatic patient.

- Oxygenation is rarely a problem and a Sp02 of >90% is acceptable
- Peak inspiratory pressure (PIP) and plateau pressure (P plat) should be kept at safe levels to minimize barotrauma
- Low tidal volume ventilation is usually necessary to avoid high PIP and P plat
- Increasing the I:E ratio by using a low initial breath rate (adults 8-10, Peds 12-16) helps minimize barotrauma
- The combination of Low VT and low breath rate results in hypercapnia which is usually acceptable in the asthmatic
- PEEP of 5-8 cmH20 may maximize oxygenation and ventilation

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6.7.5 Pediatric Asthma Score

Evaluation of the degree of respiratory distress of pediatric patients can be challenging. You may use the following scoring system to help assess pediatric patients:

6.7.6 Pediatric Respiratory Assessment Measure (PRAM)

Pediatric Respiratory Assessment Measure (PRAM)					
Signs / Scoring	**0**	**1**	**2**	**3**	**Patient Score**
Suprasternal Retractions	Absent		Present		(max 2)
Scalene muscle Contraction	Absent		Present		(max 2)
Air Entry	Normal	Decrease at bases	Widespread and Decreased	Absent / Minimal	(max 3)
Wheezing	Absent	Expiratory Only	Inspiratory and Expiratory	Audible or silent chest / no air movement	(max 3)
O2 Saturation on Room Air	≥ 95%	92-94%	< 92%		(max 2)
If there are asymmetric findings between lungs, the most severe is rated				**PRAM Score Total:**	(max 12)
PRAM 0-3 = MILD Asthma		PRAM 4-7 = MODERATE Asthma		PRAM = 8-12 SEVERE Asthma	

6.8 Diabetic Crisis

The sending physician, receiving physician, or PHI medical director must act as medical control and provide direct orders for the management of diabetic Ketoacidosis (DKA) and Hyperosmolar Hyperglycemic State (HHS). Medical crews must clarify the following components of therapy with medical control prior to transport.

6.8.1 Fluid and Electrolytes

- Clarify orders for IV fluid replacement and correcting electrolyte abnormalities.
- Normal saline should be used for the initial fluid resuscitation and treatment of cardiovascular instability.

6.8.2 Insulin

- Clarify orders for insulin therapy.
- Ideally, confirm serum potassium is ≥ 3.3 mEq/L or above the attending physician's threshold for initiating insulin.
- Verify the insulin type, dose, and route of administration

6.8.3 Airway Management

NOTICE

When intubating the patient with metabolic acidosis, whatever the cause, consideration must be made of the patient's minute ventilation and every attempt must be made to match that minute ventilation, pre, peri and post intubation.

6.8.4 Monitoring

- Neurologic examinations should be done hourly and more frequently in patients with altered mental status.
- Blood glucose must be checked every 15 minutes while the patient is receiving IV insulin.
- If the blood sugar falls by >100 mg/dl/hr (greater than 25 mg/dl from the last glucose check), any extant insulin drip should be stopped and glucose checked every 15 minutes or at least 4 times until patient turnover.

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6.9 Hypoglycemia

Serum glucose should be carefully monitored on patients receiving insulin or hypoglycemic agents, are critically ill, or have an altered level of consciousness. The following signs should prompt immediate assessment and treatment:

- Neurologic deficits, diaphoresis, or pallor
- Blood sugar less than 60 mg/dL in adult and pediatric patients
- Blood sugar less than 50 mg/dL in neonates

6.9.1 Treatment

Provide initial supportive therapy including airway management, respiratory support, and resuscitation as per the applicable guidelines and then treat hypoglycemia straight away:

Dextrose

Adult

- 10–25 gm IV over 5 min
- See concentration guide
- Repeat PRN

Pediatric

- 0.5–1 gm/kg IBW IV over 5 min
- Max single dose 25 gm
- See concentration guide
- Repeat PRN

NOTE: **= glucose < 60 gm/dL**

Dextrose Concentration Guide

Allowable dextrose concentration by age

Dextrose Concentration	Adult	Ped 8-17 yr	Ped 1-8 yr	Neo to 1 yr
0.5 gm/mL D50W	Yes	Yes	**No**	**No**
0.25 gm/mL D25W	Yes	Yes	Yes	**No**
0.1 gm/mL D10W	Yes	Yes	Yes	Yes

NOTE: **If vascular access can not be obtained or dextrose is unavailable, IM** glucagon **may be administered.**

Glucagon

Adult

- 1 mg IM
- Repeat in 10 min PRN

Pediatric

- ≥ 20 kg 1 mg IM
- < 20 kg 0.5 mg IM
- Repeat in 10 min PRN

6.9.2 Special Considerations

Ongoing assessment and Monitoring

- After symptoms resolve and glucose normalizes, glucose testing should be performed every 15 minutes until there have been 4 contiguous normal results.

6.10 Hyperkalemia

The following signs should prompt immediate treatment:

- > 5.5 mEq/L with ECG changes including loss of P waves, widening of the QRS or peaked T waves
- > 6 mEq/L, with or without ECG abnormalities

6.10.1 Treatment

NOTE: **Calcium chloride or calcium gluconate may be administered**

Calcium Chloride

Adult

- 1 gm IV over 2–10 min
- Repeat every 10 minutes PRN up to 3 times

Pediatric

- 20 mg/kg IV over 2–10 min
- Repeat in 10 minutes PRN
- Maximum single dose 1 gram

NOTE: **Faster rate of IV administration may be indicated in cardiac arrest, cardiovascular instability and crush injuries**

NOTE: **Flush the IV line after administration**

Calcium Gluconate

Adult

- 3 gm IV over 2–10 minutes
- Repeat every 10 minutes PRN up to 3 times

Pediatric

- 60 mg/kg IV over 2–10 minutes
- Repeat in 10 minutes PRN
- Maximum single dose 3 gm

NOTE: **Faster rate of IV administration may be indicated in cardiac arrest, cardiovascular instability and crush injures.**

NOTE: **Flush the IV line after administration**

Calcium Advisories

- Mixing calcium with other medications may cause a precipitate to form in the IV line
- Administer to neonates with on-line medical direction
- Rapid administration may cause bradycardia and hypotension
- Monitor patients taking digoxin for acute toxicity which can develop with calcium infusion

Sodium Bicarbonate

Adult

- 50 mEq IV over 5–10 min

Pediatric

- 1 mEq/kg IBW IV over 5–10 min
- Max single dose 50 mEq

NOTE: **May give for hyperkalemia when there is suspected or lab documented metabolic acidosis**

NOTE: **May glve for cardiac arrest due to hyperkalemia, bicarbonate is NOT indicated in cardiac arrest otherwise**

NOTE: **Give IV push in cardiac arrest**

NOTE: **May give 1 dose—additional doses require a physician order**

Insulin and Glucose

The attending physician should provide orders for the management of insulin and glucose therapy. Medical crews must clarify the following therapy with medical control prior to transport.

- Glucose dose and repeat dose parameters
- Insulin type, dose, and route of administration

Albuterol

Adult

- 5 mg in 3–5 mL via nebulizer
- Repeat every 20 minutes for 3 doses PRN

Pediatric

- Refer to adult dosing

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6.11 Respiratory Emergencies

6.11.1 Laryngotracheobronchitis (Croup)

Treatment

NOTE: **A steroid should be given before transfer**
NOTE: **An ETT 0.5–1 mm smaller than usual may be needed if the patient requires intubation**

Oxygen

- 100% FiO2
- Humidified

Epinephrine

NOTE: **Administer racemic or L-epinephrine via small volume nebulizer**

Racemic Epinephrine

Adult
- 0.25–0.5 mL of 2.25% in 3 mL NS by SVN
- Repeat every 2–4 hours

Ped
- < 15 kg
- 0.25 mL mixed in 3 mL of NS by SVN may repeat every 1-2 hours

L-epinephrine

Adult
- 5 mL of 1 mg/1 mL concentration by SVN

Ped
- 0.5 mL/kg of 1 mg/1 mL concentration
- Max dose 5 mL

NOTE: **Add the contents of the parenteral 1 mg/mL epinephrine vial or ampule undiluted into the SVN**

Steroid

Methylprednisolone

Adult
- 125 mg IV over 1 minute

Pediatric
- 2 mg/kg IBW IV over 1 minute
- Maximum single dose 125 mg

Other Acceptable Steroids

NOTE: **Give under direct medical control**

- Dexamethasone
- Prednisone

6.12 Sepsis

Sepsis is diffuse inflammatory response with life-threatening organ dysfunction caused by a dysregulated host response to infection. It leads directly to severe capillary leak, severe oxygen consumption vs. delivery deficit, and ultimately acute organ dysfunction.

When diagnosing and treating a patient with sepsis, there are 3 critical components:

- Comprehensive patient screening and early identification of sepsis using the qSOFA + EtCO2 criteria
- Aggressive source control
- Sepsis resuscitation

6.12.1 Identification and Initial Treatment

The diagnosis of sepsis is based on two mandatory criteria identified during patient assessment, infection and organ dysfunction.

Suspected or Proven Infection

- Obtain a targeted history and perform a physical exam
- Review available lab and radiologic data
- Ensure source control has been performed if possible (e.g. abscesses drained)
- Ensure broad spectrum antibiotics have been administered and continue infusing PTA antibiotics
- Implement general resuscitation measures including adequate oxygenation and airway management

NOTE: **Patients should be resuscitated prior to intubation**

Possible Sources of infection

- Encephalitis / Meningitis — Headache, nuchal symptoms, altered level of consciousness
- Pneumonia — New onset cough, rales, rhonchi, dyspnea, CXR consolidation, hypoxia
- Intra-abdominal infection — abdominal tenderness, nausea, vomiting, diarrhea
- UTI — Dysuria, flank pain, recent foley
- Cellulitis / skin infections — Erythema, pain, swelling, tissue sloughing

Establish Organ Dysfunction

Perform qSOFA + EtCO2 evaluation	Score evaluation and begin resuscitation
• RR > 22 • New altered mental status • SBP < 100 • EtCO2 < 25	• Add 1 point for each finding • > 2 points = Positive • Positive—Start 30 mL/kg bolus of NS • Negative—Reassess every 5 minutes

NOTE: **If the patient is septic collect culture reports and all source data that may be of value to the receiving facility**

6.12.2 Sepsis Resuscitation

- Complete the 30 mL/kg fluid bolus as fast as possible using a pressure bag and complete the infusion regardless of response to fluids or pressors.
- Norepinephrine may be started at any time during the fluid bolus if the *SBP < 90 or MAP < 65 mmHg*

.

Evaluate the B/P after the initial bolus is complete

Normal Saline

SBP ≥ 90 or MAP ≥65 mm/Hg?

Yes— Continue NS at 100 mL/hr

No—Administer 1 additional 20 mL/kg fluid bolus

Systematically add vasopressors to maintain SBP ≥ 90 or MAP ≥ 65 mm/Hg

Norepinephrine

- 5-70 mcg/min infusion

Once norepinephrine is > 10–20 mcg/min add

Vasopressin

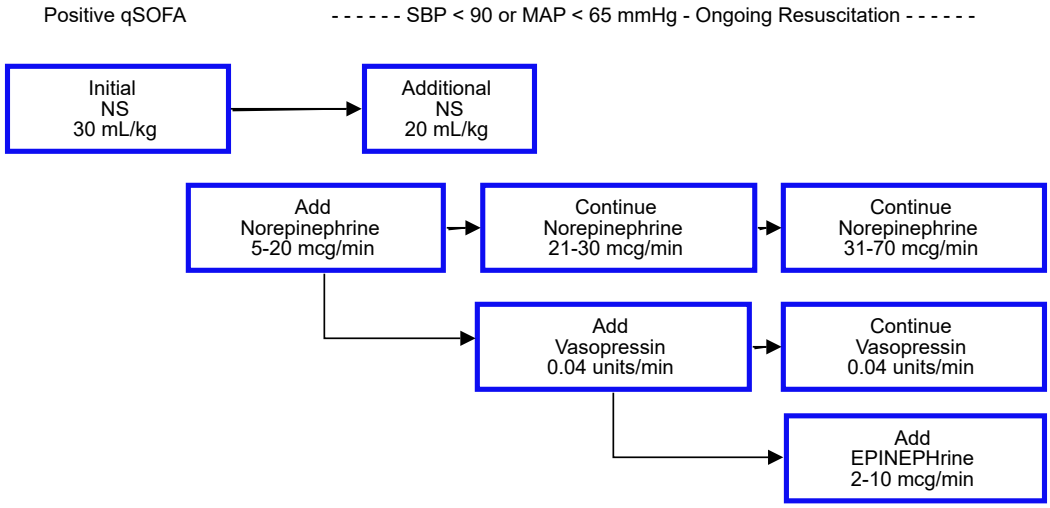
- 0.04 units/min infusion
- Do not titrate

Once norepinephrine > 30 mcg/min and vasopressin is at 0.04 units/min add

EPINEPHrine

- 2-10 mcg/min infusion

NOTE: **Norepinehrine may be increased while instituting vasopressin**
NOTE: **Titrate vasopressors down and off in reverse order**



NOTICE **Once vasopressors have started, thoroughly assess the patient for cardiogenic and hemorrhagic shock. Refer to Shock and Cardiovascular Instability.**

6.12.3 Special Considerations

- Pediatric patients with suspected sepsis should be treated under direct medical control
- Antibiotic selection and management should be done under direct medical control

Catecholamine-resistant Shock

Stress Dose Steroids

Consult Medical control if adrenal crisis is suspected and stress dose steroids are indicated.

Hydrocortisone

Adult

- 100 mg IV—usual dose
- Requires an online physician order

Pediatric

- No standing orders
- Requires an online physician order

6.13 Vascular Emergencies

6.13.1 Aortic Abdominal Aneurysm (AAA)

A well-founded diagnosis should be made by the attending physician prior to Medical Crews implementing therapy in this section.

- Obtain a detailed history of the present illness, prioritizing symptoms of abdominal pain and signs of cardiovascular instability.
- Administer oxygen as needed, adjusting the flow to maintain a SpO2 between 95% and 99%.
- Continuously monitor cardiac rhythm, oxygen saturation (SpO2), end-tidal CO2 (EtCO2), and non-invasive blood pressure (NIBP).
- Ensure the patient has a venous access line that is appropriate for rapid resuscitation and the administration of blood products if necessary.
- Review the most recent 12-lead ECG, laboratory test results, and imaging studies to inform treatment decisions.

Ruptured AAA with Hypotension and Cardiovascular Instability

NOTICE

Ruptured abdominal aortic aneurysms are acute, life-threatening conditions requiring immediate surgical intervention. These emergencies are highly time-dependent, with outcomes significantly influenced by the speed of diagnosis and surgical response.

Hemorrhage Control

- For ruptured AAA, urgent surgical repair is essential. Transport to a facility capable of emergency vascular surgery, reducing time to definitive care.
- Resuscitation Goals: Begin cautious fluid resuscitation to achieve adequate organ perfusion without exacerbating hemorrhage. Aim for a balance to avoid both over-resuscitation and under-resuscitation.

Titration of Treatment

- Radial Pulse: Use the presence of a palpable radial pulse as an indicator of adequate systolic blood pressure (SBP), adjusting treatment to maintain SBP in the 80-100 mmHg range.
- Normal Mentation: Monitor consciousness and cognitive status to ensure cerebral perfusion. Changes in mentation indicate the need for treatment adjustment.
- Manage Hypothermia, Acidosis, and Coagulopathy: Monitor and address the patient's temperature, acid-base balance, and coagulation to prevent complications that can worsen outcomes.

6.13.2 Acute Aortic Dissection

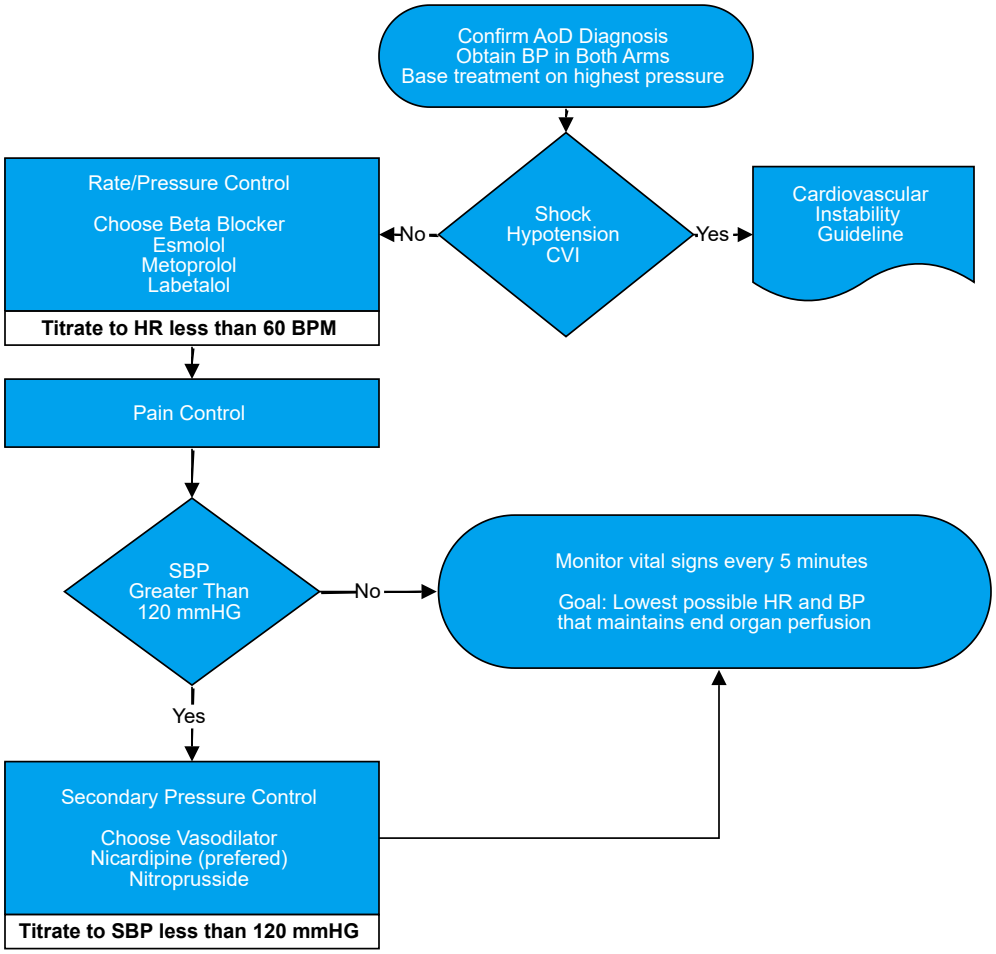
- A well-founded diagnosis should be made by the attending physician prior to implementing therapy in this section.
- Transport management is focused on controlling aortic shear stress to limit false lumen propagation and facilitating timely transport to surgical or endovascular repair.

Assessment

- Obtain a history of current illness with emphasis on chest pain, CV instability, and neurological changes
- Assess pain and ensure adequate analgesia
- Assess and compare central and peripheral pulses for deficits
- Measure and compare blood pressure in both arms
- Titrate treatment based on the higher BP of the 2 arms if there is a discrepancy
- Avoid IV lines in extremities with pulse deficits.
- Review 12 lead ECG, laboratory studies, and imaging reports

NOTICE **Type A dissections affecting the ascending aorta usually require emergent surgical treatment requiring timely preparation and transport.**

Initial Management



Initial Heart Rate, Blood Pressure and Pain Control

- Treat pain with opiod analgesics
- Administer a beta blocker to reduce the force of ventricular contraction, heart rate and blood pressure.

NOTE: **Goal: Lowest possible heart rate and systolic blood pressure that maintains end organ perfusion.**

Beta Blocker Therapy

- Target HR to less than 60 BPM
- Choose 1 of the following:

Esmolol

Adult

- Initial load: 500 mcg/kg IV over 1 min
- Titrate 50 mcg/kg/min every 5 min to target HR or BP
- Repeat the loading dose of 500 mcg/kg before every titration PRN until near target HR
- Max infusion dose 300 mcg/kg/min

Pediatric

- No PHI Standing Orders or Indications

Metoprolol—AMI, Tachyarrhythmias, Vascular Emergencies

Adult

- 5 mg slow IV push over 2–3 min
- Repeat every 5 min PRN
- Max total dose 15 mg

Pediatric

- No PHI Standing Orders or Indications

Labetalol

Adult

- 20 mg IV over 2 minutes
- Repeat 40 mg in 10-20 min PRN
- Repeat 80 mg every 10-20 min up to 3 times PRN
- Alternately 2mg/min IV infusion
- Administer to target BP. HR, or max total dose of 300 mg

NOTE: **Labetalol is a combined alpha and beta blocking agent**

Pediatric

- No PHI Standing Orders or Indications

Secondary Blood Pressure Control

- Beta blockers alone may not reduce the blood pressure to the target level.
- If the systolic blood pressure remains above 120 mmHg after pain is controlled and the heart rate is consistently <60 BPM, initiate vasodilator therapy.

NOTE: **Goal: Lowest possible heart rate and systolic blood pressure that maintains end organ perfusion.**

Vasodilator Therapy

- Target SBP 100 to 120 mm/Hg

⚠ CAUTION **Vasodilator therapy without prior beta blockade may cause reflex tachycardia and increased force of ventricular contraction leading to greater aortic wall stress and potentially causing false lumen propagation.**

Nicardipine—Preferred

Adult

- 5 mg/hr IV infusion
- Increase by 2.5 mg/hr every 5–15 minutes PRN
- Titrate to BP target
- Max 15 mg/hr

NOTE: **The dose may be reduced to 3 mg/hr after targeted response is achieved**

Ped

- No PHI Standing Orders or Indications

Nicardipine Infusion	0.1 mg/mL
Dose mg/hr	**Infusion Rate mL/hr**
3	30
5	50
7.5	75
10	100
12.5	125
15	150
Recommended Mix • 25 mg/250 mL—NS or D5W	

Clevidipine

Adult

- Initiate infusion at 1-2 mg/hr
- Dose may initially be doubled every 90 sec
- As BP approaches goal, reduce the rate of increase by less than double and lengthen time between dose adjustments to every 5–10 min.
- Max dose 32 mg/hr

Ped

- No PHI Standing Orders or Indications

Nitroprusside—Alternate

Adult

- 0.3 to 10 mcg/kg/min
- Start initial infusion at 0.3 mcg/kg/min
- Increase by 0.3 mcg/kg/min Q 5 minutes PRN
- Dose should be kept at 2 mcg/kg/min or less to avoid toxicity
- Titrate to target BP

Pediatric

- Refer to adult dosing

6.13.3 Leaking Aortic Endovascular Repair

If the SBP is greater than 120 mm/Hg, medical crews should consult with the treating surgeon, attending providers or program medical directors and manage blood pressure collaboratively. If medical direction is unavailable, medical crews may treat hypertension under the following conditions and guidelines:

Blood pressure and heart rate may be carefully treated with short-acting agents

- Target SBP 90-120mm/Hg
- Target HR 60-80 BPM

NOTE: **The following short-acting agents should be used if they are available**

⚠ CAUTION **Always be prepared to discontinue these agents and perform blood and volume resuscitation if the patient suddenly deteriorates**

Prefered Primary Antihypertensives

Esmolol

Adult

- Initial load: 500 mcg/kg IV over 1 min
- Titrate 50 mcg/kg/min every 5 min to target HR or BP
- Repeat the loading dose of 500 mcg/kg before every titration PRN until near target HR
- Max infusion dose 300 mcg/kg/min

Pediatric

- No PHI Standing Orders or Indications

Nicardipine

Adult

- 5 mg/hr IV infusion
- Increase by 2.5 mg/hr every 5–15 minutes PRN
- Titrate to BP target
- Max 15 mg/hr

NOTE: **The dose may be reduced to 3 mg/hr after targeted response is achieved**

Ped

- No PHI Standing Orders or Indications

Clevidipine

Adult

- Initiate infusion at 1-2 mg/hr
- Dose may initially be doubled every 90 sec
- As BP approaches goal, reduce the rate of increase by less than double and lengthen time between dose adjustments to every 5–10 min.
- Max dose 32 mg/hr

Ped

- No PHI Standing Orders or Indications

Secondary Antihypertensives

The following medications may be used if the preferred short-acting agents are not available

Metoprolol

Adult

- 5 mg slow IV push over 2–3 min
- Repeat every 5 min PRN
- Max total dose 15 mg

Pediatric

- No PHI Standing Orders or Indications

6.14 Nausea and Vomiting

6.14.1 Dosing and Administration

Ondansetron

Adult

- 4 mg IV over 2-5 min, IM
- Alternately 4 mg orally disintegrating tablets
- May repeat once in 30 min PRN

Pediatric

- 0.1 mg/kg IBW IV over 2-5 min
- Max single dose 4 mg
- May repeat once in 30 min PRN

Prochlorperazine

Adult

- 5–10 mg IV over 2 min, IM
- Max total dose 10 mg

Pediatric (patients 10–40 kg)

- 2.5 mg IV over 2 min, IM
- Max total dose 2.5mg
- Pataients greater than 40 kg see adult dosing
- Contraindicated in patients less than 2 years or 10 kg

NOTE: **Should be used as a second line antiemetic after ondansetron**

Promethazine

Adult

- 12.5–25 mg diluted in 10 mL IV over 2 min
- Alternately 25 mg deep IM
- May repeat q 4–6 hours PRN

Pediatric

- Age 2 years or greater
- 0.25–0.5 mg/kg diluted in 10 mL IV over 2 min
- Max single dose 12.5 mg
- Repeat q 4–6 hours PRN

NOTE: **Should be used as a second line antiemetic after ondansetron**

Promethazine—Advisories

⚠WARNING **US Boxed Warning: Promethazine should not be used in pediatric patients younger than 2 years because of the potential for fatal respiratory depression.**

⚠CAUTION **Promethazine injection can cause severe tissue injury. Check IV patency prior to each administration and monitor for medication infiltration.**

- Reduce dosage or use an alternate medication in the elderly or patients receiving concomitant sedating medications
- Refer to extrapyramidal symptoms guideline PRN
- NS, D5, or LR may be used for dilution

6.15 Extrapyramidal Symptoms

Medication induced extrapyramidal symptoms may occur after administration of some antiemetics like promethazine and prochlorperazine.

- Tremor,
- Slurred speech
- Akathesia
- Dystonia
- Anxiety
- Distress
- Paranoia
- Bradyphrenia

Medication induced extrapyramidal symptoms should be treated with diphenhydriamine.

Diphenhydramine

Adult

- 25–50 mg IV, IM
- Repeat every 4 hours PRN

Pediatric

- 1–2 mg/kg IV, IM
- Maximum single dose 50 mg
- Repeat every 4 hours PRN